

Goodness-of-Fit Tests via Entropy-Based Density Estimation Techniques

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Abstract

Goodness-of-fit testing remains a fundamental problem in statistical inference with broad practical importance. In this paper, we introduce two new goodness-of-fit tests grounded in entropy-based density estimation techniques. The first is a boundary-corrected empirical likelihood ratio test, which refines the classic approach by addressing bias near the support boundaries. The second is a novel test built on Correa's local linear entropy estimator, leveraging quantile regression to improve density estimation accuracy. We establish the theoretical properties of both test statistics and demonstrate their practical effectiveness through extensive simulation studies and real-data applications. The results show that the proposed methods deliver strong power and flexibility for assessing model adequacy in a wide range of settings.

Keywords: Bootstrap; Empirical likelihood; Goodness-of-fit test; Local linear estimator; Model

assessment; Sample entropy.

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1 Introduction

Goodness-of-fit testing is a fundamental problem in statistical inference with widespread applications in model selection, validation, and diagnostics. In the parametric framework, the classical approach is often based on the likelihood ratio test (LRT), as motivated by the Neyman–Pearson lemma. While the lemma guarantees the optimality of the LRT for testing simple hypotheses, the LRT is also widely applied to composite hypotheses due to its favorable asymptotic properties.

Let $X_1, \dots, X_n$ be independent and identically distributed (i.i.d.) observations from a continuous distribution F with density f . We consider the hypothesis-testing problem

$$H_0 : f = f_0 \quad \text{vs.} \quad H_1 : f = f_1,$$

where under the null hypothesis $f_0(x) = f_0(x; \boldsymbol{\theta})$ is known up to a parameter vector $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_d)$, and under the alternative, f_1 is completely unspecified. The classical likelihood ratio test statistic is given by

$$\Lambda = \frac{\prod_{i=1}^n f_1(X_i)}{\prod_{i=1}^n f_0(X_i)}.$$

We reject H_0 in favor of H_1 for large values of Λ , or equivalently, when the log-likelihood ratio

$$2 \log \Lambda = 2 \sum_{i=1}^n \log \left(\frac{f_1(X_i)}{f_0(X_i)} \right)$$

exceeds a critical threshold. Under standard regularity conditions and assuming H_0 holds, the distribution of $2 \log \Lambda$ converges asymptotically to a chi-squared distribution with degrees of freedom equal to the difference in dimensionality between the null and alternative models (see, e.g., Casella and Berger, 2002).

However, in many practical situations—particularly in nonparametric settings—the alternative

density f_1 is unknown, rendering direct application of the LRT infeasible. To address this, nonparametric density-based methods have been developed as flexible alternatives. Among these, empirical likelihood ratio tests and entropy-based statistics have attracted considerable attention.

Vexler and Gurevich (2010) introduced an empirical likelihood ratio test in which the unknown density f_1 is estimated nonparametrically using Vasicek's (1976) entropy estimator. They defined the empirical likelihood ratio using

$$\prod_{i=1}^n f_{1,n}(X_{(i)}),$$

where

$$f_{1,n}(X_{(i)}) = \frac{F_n(X_{(i+m)}) - F_n(X_{(i-m)})}{X_{(i+m)} - X_{(i-m)}} = \frac{2m/n}{X_{(i+m)} - X_{(i-m)}}, \quad (1)$$

and $F_n(x)$ denotes the empirical cumulative distribution function. Here, $\hat{\theta}$ is the maximum likelihood estimate under H_0 , and $X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$ are the order statistics. Boundary corrections are made such that $X_{(i-m)} = X_{(1)}$ if $i \leq m$, and $X_{(i+m)} = X_{(n)}$ if $i \geq n - m$. The integer m , called the window size, is a positive number less than $n/2$. Vexler and Gurevich (2010) proposed the test statistic

$$T_{mn} = \frac{\prod_{i=1}^n \left(\frac{2m}{n(X_{(i+m)} - X_{(i-m)})} \right)}{\prod_{i=1}^n f_0(X_i; \hat{\theta})}. \quad (2)$$

Since T_{mn} depends on m , they further suggested the minimization

$$T_{mn} = \frac{\min_{1 \leq m \leq n^\delta} \prod_{i=1}^n \left(\frac{2m}{n(X_{(i+m)} - X_{(i-m)})} \right)}{\prod_{i=1}^n f_0(X_i; \hat{\theta})}, \quad (3)$$

where $\delta \in (0, 1)$.

This general methodology has since been extended to various settings, including logistic distributions (Alizadeh Noughabi, 2015), skew normality (Ning and Ngunkeng, 2013), Laplace distributions (Alizadeh Noughabi, 2016), and Rayleigh distributions (Safavinejad et al., 2014). Although the method of Vexler and Gurevich (2010) is generally effective, it is known to suffer from boundary bias, particularly near the endpoints—a limitation also discussed by Ebrahimi et al. (1994) in the

context of entropy estimation.

To address these shortcomings, we propose two test statistics. The first is a corrected version of the Vexler–Gurevich statistic (2) and (3), incorporating a position-dependent correction factor to properly account for boundary effects. The second is a new test statistic based on Correa’s (1995) local linear entropy estimator, which improves density estimation by locally interpolating the quantile function. Together, these methods aim to enhance the performance and reliability of goodness-of-fit testing.

The remainder of the paper is organized as follows. Section 2 introduces the two proposed test statistics. Section 3 presents their theoretical properties. Section 4 describes the computational implementation using a bootstrap procedure. Section 5 provides simulation studies and real-data applications to evaluate the tests’ performance. Finally, Section 6 concludes and outlines potential future research directions.

2 Two Approaches for Goodness-of-Fit Testing

In this section, we introduce two test statistics for assessing model adequacy. Both are constructed as likelihood-type ratios, where the numerator provides a nonparametric estimate of the true density—either via the boundary-corrected m -spacing method or Correa’s local linear entropy estimator—and the denominator corresponds to the fitted parametric model under the null hypothesis. Values of the statistic close to one indicate good agreement between the empirical and theoretical densities, while large values signal model misspecification.

The first test is a boundary-corrected version of the empirical likelihood ratio statistic of Vexler and Gurevich (2010), designed to mitigate bias near the support boundaries. The second test is based on Correa’s (1995) local linear entropy estimator, which improves density estimation by locally approximating the derivative of the quantile function.

2.1 Corrected Test Statistic of Vexler and Gurevich (2010)

As noted in Section 1, the original test statistic proposed by Vexler and Gurevich is subject to boundary bias, particularly when the index i lies near the extremes of the support. This bias arises because the slope between $X_{(i-m)}$ and $X_{(i+m)}$ in the empirical distribution is not accurately estimated at the boundaries. To address this issue, we propose a boundary-corrected estimator, following Ebrahimi et al. (1994) and Al-Labadi et al. (2025). The estimator is defined as

$$f_{1,n}(X_{(i)}) = \frac{F_n(X_{(i+m)}) - F_n(X_{(i-m)})}{X_{(i+m)} - X_{(i-m)}} = \frac{c_i m/n}{X_{(i+m)} - X_{(i-m)}},$$

where $X_{(i-m)} = X_{(1)}$ if $i \leq m$, and $X_{(i+m)} = X_{(n)}$ if $i \geq n - m$. The correction factor c_i is defined as

$$c_i = \begin{cases} \frac{m+i-1}{m}, & \text{if } 1 \leq i \leq m, \\ 2, & \text{if } m+1 \leq i \leq n-m, \\ \frac{n+m-i}{m}, & \text{if } n-m+1 \leq i \leq n. \end{cases}$$

This position-dependent factor replaces the constant value 2 used in the original formulation and leads to improved accuracy in density estimation near the boundaries. The resulting test statistic takes the form

$$T_{mn}^* = \frac{\prod_{i=1}^n \left(\frac{c_i m}{n(X_{(i+m)} - X_{(i-m)})} \right)}{\prod_{i=1}^n f_0(X_i; \hat{\theta})}, \quad (4)$$

where $\hat{\theta}$ denotes the maximum likelihood estimator under the null model f_0 . The selection method for m is described in (7).

2.2 Correa's Entropy-Based Density Estimator

Recognizing that the test statistic in (4) is constructed as a likelihood-type ratio, where the numerator represents a nonparametric estimate of the true density, we propose a new test statistic based on the entropy estimator of Correa (1995), which relies on estimating the derivative of the

quantile function via local linear regression. Unlike spacing-based methods, Correa's estimator directly models quantile derivatives via local linear regression, avoiding boundary issues and adaptively smoothing noise, particularly advantageous for multimodal or skewed densities. Let X be a continuous random variable with cumulative distribution function F and density f , and define the quantile function as $q(p) = F^{-1}(p)$. The Shannon entropy can be expressed in terms of the quantile function as

$$H(f) = \int_0^1 \log q'(p) dp,$$

so that entropy estimation reduces to estimating $q'(p)$. Correa's method estimates this derivative locally using linear regression.

Given a sample $X_1, \dots, X_n$, let $X_{(1)} \leq \dots \leq X_{(n)}$ denote the order statistics and define the empirical quantiles $p_i = i/n$. Since $X_{(i)} \approx F^{-1}(p_i)$, a local linear regression is performed in a neighborhood of each p_i :

$$X_{(j)} = a_i + b_i p_j + \varepsilon_j, \quad j = i - m, \dots, i + m,$$

where the slope b_i is estimated via least squares as

$$b_i = \frac{\sum_{j=i-m}^{i+m} (p_j - \bar{p}_i)(X_{(j)} - \bar{X}_i)}{\sum_{j=i-m}^{i+m} (p_j - \bar{p}_i)^2}, \quad (5)$$

with $\bar{p}_i$ and $\bar{X}_i$ denoting local means. According to Correa (1995), an estimator for $H(f)$ is then

$$\hat{H}(f) = \frac{1}{n} \sum_{i=1}^n \log b_i.$$

Since $q'(p_i) \approx b_i \approx 1/f(X_{(i)})$, the corresponding density estimate is defined as

$$f_{1,n}(X_{(i)}) = \frac{1}{b_i}.$$

The proposed test statistic based on Correa's estimator is therefore given by

$$T_{mn}^{\text{new}} = \frac{\prod_{i=1}^n f_{1,n}(X_{(i)})}{\prod_{i=1}^n f_0(X_i; \hat{\theta})} = \frac{\prod_{i=1}^n b_i^{-1}}{\prod_{i=1}^n f_0(X_i; \hat{\theta})}, \quad (6)$$

where b_i is as defined in (5). In Section 4, we demonstrate that use of (6) yields a powerful goodness-of-fit test.

3 Theoretical Results

In this section, we investigate the asymptotic properties of the proposed test statistic T_{mn}^{new} introduced in the previous section. In particular, we show that the normalized logarithm of the test statistic (6) converges to zero under the null hypothesis and to a strictly positive limit under fixed alternatives. These results hold under mild regularity conditions and do not require asymptotic normality or parametric assumptions. Instead, they follow from standard consistency properties of nonparametric density estimators and laws of large numbers.

Let $X_1, \dots, X_n$ be i.i.d. observations from a continuous distribution with density f , and let $f_0(x) = f_0(x; \theta)$ denote the fully specified null density. The test statistic T_{mn}^{new} is as defined in (6). We impose the following regularity conditions:

(C1) The density f is continuous and strictly positive on its support.

(C2) The null density f_0 is continuous and strictly positive on the support of f , and $\log f_0(x; \theta)$ is integrable under f .

It is well known that, under (C1), the local linear estimator $1/b_i$ is consistent in probability for $f(X_{(i)})$ at each fixed i (Correa, 1995); that is,

$$1/b_i \xrightarrow{P} f(X_{(i)}) \quad \text{as } n \rightarrow \infty.$$

Lemma 1 *Under conditions (C1)–(C2), the following hold:*

(i) Under $H_0 : f = f_0$, we have

$$\frac{1}{n} \log T_{mn}^{\text{new}} \xrightarrow{P} 0.$$

(ii) Under $H_1 : f \neq f_0$, we have

$$\frac{1}{n} \log T_{mn}^{\text{new}} \xrightarrow{P} \mathbb{E}_f \left[\log \left(\frac{f(X)}{f_0(X)} \right) \right] > 0,$$

where the expectation is taken with respect to the true density f .

Proof. We express the log of the test statistic as

$$\log T_{mn}^{\text{new}} = \sum_{i=1}^n (\log f_{1,n}(X_{(i)}) - \log f_0(X_i)) = \sum_{i=1}^n \left(\log \left(\frac{1}{b_i} \right) - \log f_0(X_i) \right).$$

Thus, the normalized log statistic becomes

$$\frac{1}{n} \log T_{mn}^{\text{new}} = \frac{1}{n} \sum_{i=1}^n \left(\log \left(\frac{1}{b_i} \right) - \log f_0(X_i) \right).$$

Part (i): Under the null hypothesis $f = f_0$, by the consistency of the estimator, $1/b_i \xrightarrow{P} f_0(X_i)$ for each i . By the continuous mapping theorem, $\log(1/b_i) \xrightarrow{P} \log f_0(X_i)$, so each summand converges in probability to zero. Since $\log f_0(X_i)$ is integrable by (C2), uniform integrability yields

$$\frac{1}{n} \log T_{mn}^{\text{new}} \xrightarrow{P} 0.$$

Part (ii): Under the alternative $f \neq f_0$, $1/b_i \xrightarrow{P} f(X_i)$, so

$$\log \left(\frac{1}{b_i f_0(X_i)} \right) \xrightarrow{P} \log \left(\frac{f(X_i)}{f_0(X_i)} \right).$$

The weak law of large numbers then gives

$$\frac{1}{n} \sum_{i=1}^n \log \left(\frac{1}{b_i f_0(X_i)} \right) \xrightarrow{P} \mathbb{E}_f \left[\log \left(\frac{f(X)}{f_0(X)} \right) \right].$$

The expectation on the right is the Kullback–Leibler divergence $D_{\text{KL}}(f||f_0)$, which is strictly positive under H_1 . This completes the proof. ■

This result confirms that the proposed test statistic T_{mn}^{new} is consistent and does not require knowledge of the null distribution’s asymptotic form or any parametric estimation procedure under the alternative.

Remark. The same arguments and consistency results hold for the boundary-corrected m -spacing test statistic T_{mn}^* defined in equation (4). The boundary correction, as in Ebrahimi et al. (1994), ensures that the m -spacing estimator is consistent in probability for $f(X_{(i)})$ at each sample point, even near the endpoints of the support. Thus, the asymptotic properties established above for T_{mn}^{new} also apply to T_{mn}^* under the same regularity conditions.

Unlike classical likelihood ratio tests, which rely on parameter estimation under both the null and alternative models, the proposed statistic T_{mn}^{new} depends only on parameter estimation under the null hypothesis (if required). In fully specified null cases, it avoids parameter estimation entirely. This feature simplifies implementation and enhances robustness, especially in settings where maximum likelihood estimation is computationally challenging or unreliable. This simplification enhances robustness and broadens applicability, especially in settings where maximum likelihood estimation is unstable or computationally intensive.

4 Computational Algorithm

To implement the proposed test statistics T_{mn}^* and T_{mn}^{new} , it is necessary to select an appropriate window size parameter m . A commonly used rule, suggested by Grzegorzewski and Wieczorkowski (1999), is

$$m = \lfloor \sqrt{n} + 0.5 \rfloor, \quad (7)$$

where $\lfloor \cdot \rfloor$ denotes the floor function. This choice effectively balances bias and variance in the entropy-based density estimators and is widely adopted in practice.

After computing the test statistic, the next step is to assess whether its value provides sufficient evidence to reject the null hypothesis. Values of the test statistic close to one indicate agreement between the empirical and theoretical densities, whereas large values suggest model misspecification and provide evidence against the null. Since the asymptotic distribution of T_{mn}^{new} is analytically intractable, we employ a bootstrap procedure to approximate its null distribution and obtain the corresponding p -value.

The following algorithm outlines the steps for conducting the bootstrap-based goodness-of-fit test:

1. Given an observed sample $X_1, \dots, X_n$, compute the test statistic T_{mn}^{new} as defined in equation (6).
2. Fit the null model $f_0(x; \hat{\theta})$ to the data, where $\hat{\theta}$ denotes the maximum likelihood estimator under H_0 .
3. Generate B bootstrap samples $\{X_1^{*(b)}, \dots, X_n^{*(b)}\}$, for $b = 1, \dots, B$, by sampling from the fitted null model $f_0(x; \hat{\theta})$.
4. For each bootstrap sample, compute the corresponding test statistic $T_{mn}^{\text{new},*(b)}$.
5. Estimate the bootstrap p -value as

$$p_{\text{boot}} = \frac{1}{B} \sum_{b=1}^B \mathbb{I} \left(T_{mn}^{\text{new},*(b)} \geq T_{mn}^{\text{new}} \right),$$

where $\mathbb{I}(\cdot)$ denotes the indicator function.

6. Reject the null hypothesis H_0 at significance level α if $p_{\text{boot}} < \alpha$.

This resampling approach avoids reliance on the asymptotic distribution of the test statistic, making the procedure suitable for small to moderate sample sizes and for complex models. An analogous bootstrap algorithm is used for the statistic T_{mn}^* defined in equation (4).

5 Simulation Study

In this section, we evaluate the performance of the proposed test statistics through both simulation studies and real data applications. Due to the complexity and nonparametric nature of the estimators, the exact sampling distributions under the null hypothesis are analytically intractable. Therefore, we employ the bootstrap procedure described in Section 4 to approximate the null distribution and compute the corresponding p -values.

For each scenario in Examples 1 and 2, samples of size $n \in \{20, 50, 100, 200\}$ are generated from the specified true distributions, as detailed in Tables 1 and 2. The test statistics $T_{mn}^{\text{origin}} = T_{mn}$, T_{mn}^* , and T_{mn}^{new} , as defined in equations (2), (4), and (6), respectively, are computed for each sample. Corresponding p -values are then estimated using $B = 1000$ bootstrap replications from the fitted null model. To ensure reproducibility, the random seed is set via `set.seed(2025)`.

Example 1 (*Testing $\mathcal{N}(\mu, 1)$ as the Null*): We begin by testing the null hypothesis $H_0 : f = f_0$, where f_0 is the density of $\mathcal{N}(\mu, 1)$ with μ unknown and estimated by $\hat{\mu} = \bar{x}$ for each sample.

Samples are generated from various true distributions, as listed in Table 1, to evaluate the sensitivity of the proposed test statistics to departures from normality. These alternatives include normal distributions with different means and variances, a symmetric mixture, as well as heavy-tailed and skewed distributions.

Table 1 displays the resulting p -values. When the data are truly normal—whether matching the null mean ($\mathcal{N}(0, 1)$) or with a shifted mean ($\mathcal{N}(5, 1)$)—the tests do not reject the null hypothesis and p -values are large for all sample sizes, as expected. As the alternative distributions deviate further from normality (e.g., increased variance or heavier tails), the tests become more sensitive. For instance, under the t_3 distribution, the null is not rejected for small samples ($n = 20$), but is rejected for larger n , reflecting the increase in power. For non-normal alternatives like the Cauchy and exponential distributions, the tests exhibit high power, yielding near-zero p -values even for moderate sample sizes.

Notably, T_{mn}^* and T_{mn}^{origin} yield identical results, while T_{mn}^{new} can be marginally more sensitive in some cases, especially for moderate sample sizes or challenging alternatives. Overall, the results

demonstrate that the proposed methods maintain the correct Type I error rate under the null and reliably detect departures from normality as the sample size grows.

Table 1: Bootstrap p -values (p_{boot}) for various true distributions under $H_0 : f = f_0$, where f_0 is the density of $\mathcal{N}(\mu, 1)$, using T_{mn}^* , T_{mn}^{new} , and T_{mn}^{origin} .

Sample Size	True Distribution	T_{mn}^*	T_{mn}^{new}	T_{mn}^{origin}
20	$\mathcal{N}(0, 1)$	0.995	0.949	0.995
	$\mathcal{N}(5, 1)$	0.992	0.927	0.992
	$\mathcal{N}(0, 4)$	0.002	0.000	0.002
	$0.5\mathcal{N}(-1, 1) + 0.5\mathcal{N}(1, 1)$	0.006	0.010	0.006
	t_3	0.663	0.571	0.663
	Logistic(0,1)	0.131	0.076	0.131
	Cauchy(0,1)	0.000	0.000	0.000
	Exponential(1)	0.034	0.031	0.034
50	$\mathcal{N}(0, 1)$	0.757	0.745	0.757
	$\mathcal{N}(5, 1)$	0.776	0.751	0.776
	$\mathcal{N}(0, 4)$	0.000	0.000	0.000
	$0.5\mathcal{N}(-1, 1) + 0.5\mathcal{N}(1, 1)$	0.005	0.000	0.005
	t_3	0.002	0.000	0.002
	Logistic(0,1)	0.000	0.000	0.000
	Cauchy(0,1)	0.000	0.000	0.000
	Exponential(1)	0.000	0.000	0.000
100	$\mathcal{N}(0, 1)$	0.917	0.958	0.917
	$\mathcal{N}(5, 1)$	0.928	0.965	0.928
	$\mathcal{N}(0, 4)$	0.000	0.000	0.000
	$0.5\mathcal{N}(-1, 1) + 0.5\mathcal{N}(1, 1)$	0.000	0.000	0.000
	t_3	0.000	0.000	0.000
	Logistic(0,1)	0.000	0.000	0.000
	Cauchy(0,1)	0.000	0.000	0.000
	Exponential(1)	0.000	0.000	0.000
200	$\mathcal{N}(0, 1)$	0.737	0.530	0.737
	$\mathcal{N}(5, 1)$	0.740	0.541	0.740
	$\mathcal{N}(0, 4)$	0.000	0.000	0.000
	$0.5\mathcal{N}(-1, 1) + 0.5\mathcal{N}(1, 1)$	0.000	0.000	0.000
	t_3	0.000	0.000	0.000
	Logistic(0,1)	0.000	0.000	0.000
	Cauchy(0,1)	0.000	0.000	0.000
	Exponential(1)	0.000	0.000	0.000

Example 2 (Testing $\mathcal{N}(\mu, \sigma^2)$ as the Null): In this simulation, we test the null hypothesis $H_0 : f = f_0$, where f_0 is the density of the normal distribution $\mathcal{N}(\mu, \sigma^2)$ with both mean μ and

variance σ^2 unknown. The maximum likelihood estimators $\hat{\mu} = \bar{x}$ and $\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$ are used to fit the null model.

Samples are generated from the same set of alternative distributions as in Example 1, including normal distributions with different means and variances, symmetric mixtures, as well as heavy-tailed and skewed distributions. For each sample size $n \in \{20, 50, 100, 200\}$, the test statistics T_{mn}^* , T_{mn}^{new} , and T_{mn}^{origin} are computed, and bootstrap p -values are estimated using $B = 1000$ replications.

Table 2 reports the resulting p -values, which provide insight into the performance of the tests when both location and scale are unknown. When the data are drawn from a normal distribution (either with the same or shifted mean), the tests maintain the nominal Type I error rate, as p -values remain large across all sample sizes. For alternatives with heavier tails, skewness, or mixtures, the p -values decrease rapidly as the sample size increases, demonstrating the ability of the proposed tests to detect deviations from the null model. The tests are especially powerful against the Cauchy and exponential alternatives, with p -values near zero even for small samples. When both location and scale are unknown, the tests show some loss of sensitivity to changes in variance alone, as seen in the $N(0, 4)$ case, where p -values only decrease substantially for larger samples. Overall, T_{mn}^* and T_{mn}^{origin} produce nearly identical results, while T_{mn}^{new} can exhibit greater sensitivity in challenging cases and with moderate sample sizes. These results highlight the robustness and power of the proposed methods for model assessment under a composite normal null hypothesis.

Example 3 (*Real Data – Yarn Strength*): We apply the proposed tests to the breaking strength

Table 2: Bootstrap p -values (p_{boot}) for various true distributions under $H_0 : f = f_0$, where f_0 is the density of the normal distribution $\mathcal{N}(\mu, \sigma^2)$, using T_{mn}^* , T_{mn}^{new} , and T_{mn}^{origin} .

Sample Size	True Distribution	T_{mn}^*	T_{mn}^{new}	T_{mn}^{origin}
20	$\mathcal{N}(0, 1)$	0.999	0.986	0.999
	$\mathcal{N}(5, 1)$	0.998	0.978	0.998
	$\mathcal{N}(0, 4)$	0.524	0.411	0.524
	$0.5\mathcal{N}(-1, 1) + 0.5\mathcal{N}(1, 1)$	0.605	0.795	0.605
	t_3	0.983	0.992	0.983
	Logistic(0,1)	0.936	0.888	0.936
	Cauchy(0,1)	0.000	0.000	0.000
	Exponential(1)	0.047	0.039	0.047
50	$\mathcal{N}(0, 1)$	0.767	0.745	0.760
	$\mathcal{N}(5, 1)$	0.774	0.773	0.774
	$\mathcal{N}(0, 4)$	0.765	0.744	0.765
	$0.5\mathcal{N}(-1, 1) + 0.5\mathcal{N}(1, 1)$	0.424	0.676	0.424
	t_3	0.891	0.736	0.891
	Logistic(0,1)	0.899	0.641	0.899
	Cauchy(0,1)	0.000	0.000	0.000
	Exponential(1)	0.000	0.000	0.000
100	$\mathcal{N}(0, 1)$	0.919	0.961	0.919
	$\mathcal{N}(5, 1)$	0.928	0.959	0.928
	$\mathcal{N}(0, 4)$	0.930	0.965	0.930
	$0.5\mathcal{N}(-1, 1) + 0.5\mathcal{N}(1, 1)$	0.000	0.000	0.000
	t_3	0.000	0.000	0.000
	Logistic(0,1)	0.722	0.481	0.722
	Cauchy(0,1)	0.000	0.000	0.000
	Exponential(1)	0.000	0.000	0.000
200	$\mathcal{N}(0, 1)$	0.756	0.543	0.756
	$\mathcal{N}(5, 1)$	0.709	0.533	0.709
	$\mathcal{N}(0, 4)$	0.733	0.518	0.733
	$0.5\mathcal{N}(-1, 1) + 0.5\mathcal{N}(1, 1)$	0.030	0.001	0.030
	t_3	0.000	0.000	0.000
	Logistic(0,1)	0.939	0.952	0.939
	Cauchy(0,1)	0.000	0.000	0.000
	Exponential(1)	0.000	0.000	0.000

values of 100 yarns, originally reported by Duncan (1974):

66, 117, 132, 111, 107, 85, 89, 79, 91, 97, 138, 103, 111,
86, 78, 96, 93, 101, 102, 110, 95, 96, 88, 122, 115, 92,
137, 91, 84, 96, 97, 100, 105, 104, 137, 80, 104, 104, 106,
84, 92, 86, 104, 132, 94, 99, 102, 101, 104, 107, 99, 85,
95, 89, 102, 100, 98, 97, 104, 114, 111, 98, 99, 102, 91,
95, 111, 104, 97, 98, 102, 109, 88, 91, 103, 94, 105, 103,
96, 100, 101, 98, 97, 97, 101, 102, 98, 94, 100, 98, 99,
92, 102, 87, 99, 62, 92, 100, 96, 98

We assess whether these data are adequately modeled by a Laplace distribution,

$$f(x) = \frac{1}{2\sigma} \exp\left(-\frac{|x - \mu|}{\sigma}\right),$$

where μ and $\sigma > 0$ denote the location and scale parameters, respectively. The maximum likelihood estimates are $\hat{\mu} = 99$ and $\hat{\sigma} = 8.33$, consistent with the values reported by Alizadeh Noughabi (2016). Using these estimates, we compute the p -values for T_{mn}^{origin} , T_{mn}^* , and T_{mn}^{new} via a bootstrap procedure with $B = 1000$ replications. The resulting p -values, 0.493, 0.493, and 0.875, are all well above the conventional 0.05 significance level, providing no evidence against the Laplace model. These results support the adequacy of the Laplace distribution for the yarn strength data.

Example 4 (Real Data – River Flow): We assess the suitability of the three-parameter gamma distribution for modeling river flow measurements. The density is given by

$$f(x; \alpha, \beta, \gamma) = \frac{\beta^\alpha (x - \gamma)^{\alpha-1} e^{-\beta(x-\gamma)}}{\Gamma(\alpha)}, \quad x > \gamma,$$

where $\alpha > 0$, $\beta > 0$, and $\gamma > 0$ are the shape, rate, and location parameters, respectively.

The dataset comprises river flow measurements (in millions of cubic feet per second) from the Susquehanna River at Harrisburg, Pennsylvania, recorded over the five-year period 1980–1984:

0.654, 0.613, 0.315, 0.449, 0.297, 0.402, 0.379, 0.423, 0.379, 0.324,
0.269, 0.740, 0.418, 0.412, 0.494, 0.416, 0.338, 0.392, 0.484, 0.265

Following Al-Labadi and Evans (2018), the maximum likelihood estimates for the parameters are $\hat{\alpha} = 1.7050$, $\hat{\beta} = 0.0955$, and $\hat{\gamma} = 0.2602$. Using these estimates, we apply the bootstrap algorithm with $B = 1000$ replications to compute the p -values for the T_{mn}^{origin} , T_{mn}^* , and T_{mn}^{new} test statistics.

The resulting p -values are 0.810 for T_{mn}^{origin} and T_{mn}^* , and 0.916 for T_{mn}^{new} . These values are well above conventional significance thresholds, indicating strong agreement between the observed data and the fitted three-parameter gamma model. This demonstrates that the proposed tests do not

spuriously reject an adequate model and confirms the suitability of the gamma distribution for describing the river flow data in this context. These results are consistent with the findings of the Bayesian nonparametric test proposed by Al-Labadi and Evans (2018).

To further evaluate the effectiveness of the proposed tests in detecting model misspecification, we conduct a power analysis under several alternative distributions, as well as under the null hypothesis. For each scenario, we fix the significance level at $\alpha = 0.05$ and generate 1000 independent samples for each sample size $n \in \{20, 50, 100, 200\}$.

For each sample, the test statistics T_{mn}^{origin} , T_{mn}^* and T_{mn}^{new} are computed, and the corresponding p -values are estimated using $B = 1000$ bootstrap replications from the null model $\mathcal{N}(0, 1)$, as described in Section 4. The empirical rejection rate is calculated as the proportion of samples for which the null hypothesis is rejected.

To assess Type I error control, we also report results under the null hypothesis H_0 , where samples are drawn from $\mathcal{N}(0, 1)$. The empirical rejection rates in this case should be close to the nominal level α , indicating the validity of the bootstrap calibration and the reliability of the testing procedure.

The alternative distributions considered highlight various types of departures from the null, including a mean shift ($\mathcal{N}(1, 1)$), heavy tails ($\text{Cauchy}(0, 1)$), and a symmetric distribution with different kurtosis ($\text{Logistic}(0, 1)$). This range of alternatives enables a comprehensive assessment of the sensitivity and robustness of the proposed methods.

The results, presented in Table 3, demonstrate that both tests maintain appropriate Type I error rates when H_0 is true. Moreover, the power increases rapidly with sample size and is highest when the true distribution deviates substantially from the null. Notably, the test based on Correa's local linear entropy estimator (T_{mn}^{new}) consistently outperforms the boundary-corrected m -spacing test (T_{mn}^*), particularly for small to moderate samples and challenging alternatives. These findings confirm the strong performance of the proposed methods, both in maintaining control of Type I error and in delivering high power against a range of alternatives.

Taken together, the simulation results and real-data examples demonstrate that both proposed tests provide reliable inference for model adequacy across a range of scenarios. The tests maintain

Table 3: Empirical rejection rates at significance level $\alpha = 0.05$ based on 1000 simulations.

Alternative Distribution	Sample Size	T_{mn}^*	T_{mn}^{new}	T_{mn}^{origin}
Normal(0,1) (Null)	20	0.041	0.044	0.041
	50	0.049	0.041	0.049
	100	0.048	0.045	0.048
	200	0.056	0.049	0.056
Normal(1,1)	20	0.942	0.953	0.942
	50	1.000	1.000	1.000
	100	1.000	1.000	1.000
	200	1.000	1.000	1.000
Cauchy(0,1)	20	0.980	0.983	0.980
	50	1.000	1.000	1.000
	100	1.000	1.000	1.000
	200	1.000	1.000	1.000
Logistic(0,1)	20	0.830	0.846	0.830
	50	0.977	0.991	0.977
	100	1.000	1.000	1.000
	200	1.000	1.000	1.000

proper Type I error control under the null, deliver substantial power under diverse alternatives, and do not spuriously reject in well-specified real-data settings. The new test based on Correa’s entropy estimator in particular offers notable advantages for moderate samples and challenging alternatives, while the boundary-corrected m -spacing approach retains strong and consistent performance. These findings illustrate the utility and flexibility of entropy-based density estimation methods for modern goodness-of-fit testing.

6 Conclusion

In this paper, we proposed two bootstrap-based test statistics for assessing the goodness-of-fit of fully specified parametric models. The first test is a boundary-corrected version of the empirical likelihood ratio statistic originally introduced by Vexler and Gurevich (2010), which incorporates a position-dependent correction factor to improve density estimation near the boundaries. The second test is based on Correa’s (1995) local linear entropy estimator, which provides a flexible and accurate alternative by utilizing local linear regression to approximate the derivative of the quantile

function.

We established the theoretical properties of the proposed statistics, demonstrating their consistency and showing that, under fixed alternatives, they converge to the Kullback–Leibler divergence. Since the asymptotic distributions of these statistics are analytically intractable, we developed a bootstrap algorithm for practical implementation.

Comprehensive simulation studies demonstrated that both test statistics perform well in terms of controlling Type I error and detecting model misspecification. In particular, the test based on Correa’s estimator exhibited superior power across a wide range of alternatives, especially in small to moderate sample sizes. Applications to real data further confirmed the utility and flexibility of the proposed methods.

Unlike classical likelihood ratio tests, which require parameter estimation under both the null and alternative models, the proposed approach only requires parameter estimation under the null hypothesis (if at all). In the case of a fully specified null, parameter estimation is entirely avoided. This significantly simplifies implementation and enhances robustness, especially in settings where maximum likelihood estimation is computationally challenging or unreliable.

Future research directions include extending these tests to multivariate distributions, regression models, and right-censored data.

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